with in everyday life, but in the very early universe, at very high energies, they were aspects of the same thing. The Higgs field was instrumental in that transition; when it changed, the laws of physics changed too.

This is a big part of why we build colliders: to create, in tiny little spaces inside our detectors, the kinds of extreme conditions that existed at the beginning of the universe and that can give us insights into the underlying physical principles that dictate how *everything* in physics fits together. The basic idea is that there must be some kind of overarching mathematical theory that gives us a blueprint for particle interactions under all possible conditions, and by continually producing higher and higher energy interactions, we get an increasingly clear picture of what that larger framework looks like.

As an analogy, think of water. At the most fundamental level, it's a collection of molecules made up of bonded hydrogen and oxygen atoms in a particular arrangement. But our everyday experience of water is as a uniform colorless liquid, or perhaps as a crystalline solid, or at certain unfortunate times as the kind of soul-crushing humidity that makes you wish your clothing were made of towels.[fn5] By examining water's behavior in these different forms, we can make inferences about what it *really is*, even if we don't have powerful microscopes at hand to see the individual atoms themselves. The shape of a snowflake, for example, tells us something about the shape of the molecules as they arrange into crystals. The way water evaporates tells us something about the bonds that hold the molecules together. If we only ever experienced water in one of its phases, we wouldn't have a full picture of it, and it would be harder to get to that complete story. In the same way, our experience of subatomic particle interactions changes based on the energy (or temperature) of the experiment, and that allows us to get a better view of what's really going on.

What we want to know in particle physics is how the particles interact with each other and how their fundamental properties, like their masses, came to be what they are. The salient feature of any particle that has mass is that it can't accelerate without an application of force, and it can never reach the speed of light. In the very early universe, the Higgs field underwent a transition that separated the electroweak force into electromagnetism and the weak force, and in the process gave some particles (though not the photon or gluon) the ability to interact with the